The Value of Power Flexibility: Battery Day-Ahead Arbitrage, France vs Germany (2019–2025)

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Why does a battery earn less in France than in Germany, and how has the gap evolved since 2019?

1 How European power markets work

In Europe, electricity can be exchanged and traded between the different participating countries, whose markets are coupled through interconnectors. This helps find a balance between supply and demand. For example, if a country is producing too much electricity for its domestic demand, the excess flows to neighbouring countries where it is more expensive, within the limits of cross-border capacity. Electricity can be traded on different markets. First, the futures market is mainly used to protect energy suppliers and big industrial consumers against volatility or price peaks, but traders also use it to take positions.

The spot (day-ahead) market sets the energy price for the day after, hour by hour. This is made to adjust almost perfectly the balance between the planned production of electricity and the estimated demand for the next day. This is the most interesting market for this project, since the battery charges and discharges based on these prices.

Finally, the last market is the intraday market. Here, last-minute unexpected events are corrected up to a few minutes before delivery, to ensure that a change in the weather or the breakdown of a plant won't have a major impact on the system. This market is not modelled in this project.

1.1 The merit order

The day-ahead price of electricity is determined by the merit order [2]: plants are dispatched from the lowest to the highest marginal cost (variable cost) until demand is met, and the price is set by the last plant needed, the marginal plant. All producers then receive this price. By definition, the marginal cost of solar electricity is very low, as solar panels have almost no operating costs. The merit order typically ranks generation technologies as follows, although the exact order varies with fuel and carbon prices:

1. Solar – near-zero marginal cost (no fuel)
2. Wind (onshore and offshore) – near-zero marginal cost
3. Run-of-river hydro – near-zero marginal cost, no storage
4. Nuclear – low marginal cost, high fixed costs
5. Biomass – fuel cost, often subsidised
6. Lignite – very cheap fuel but high CO_2 emissions ($\sim$1.1 t/MWh)

7. Hard coal – imported fuel + CO_2 (∼0.9 t/MWh)
8. Gas, combined cycle (CCGT) – gas price + CO_2 (∼0.37 t/MWh)
9. Gas, open cycle (OCGT peakers) – low efficiency (∼35–40%)
10. Oil / diesel – most expensive, rarely dispatched

Coal and gas can switch places depending on the gas-to-carbon price ratio (coal-to-gas switching). Reservoir hydro has no fixed position: it bids at its opportunity cost, often just below gas.

1.2 Why batteries can arbitrage

As a consequence of this ranking, a sunny day is generally synonymous with low electricity prices around midday in countries where solar energy is well developed. Yet, in the evening, when the sun sets and demand peaks, solar panels no longer produce and the price relies on more expensive technologies such as gas. This is where an arbitrage opportunity appears thanks to batteries: buy when prices are low (around midday, for example) and sell when prices get higher (during the evening peak, for example), within the same day.

1.3 A first look: one sunny day

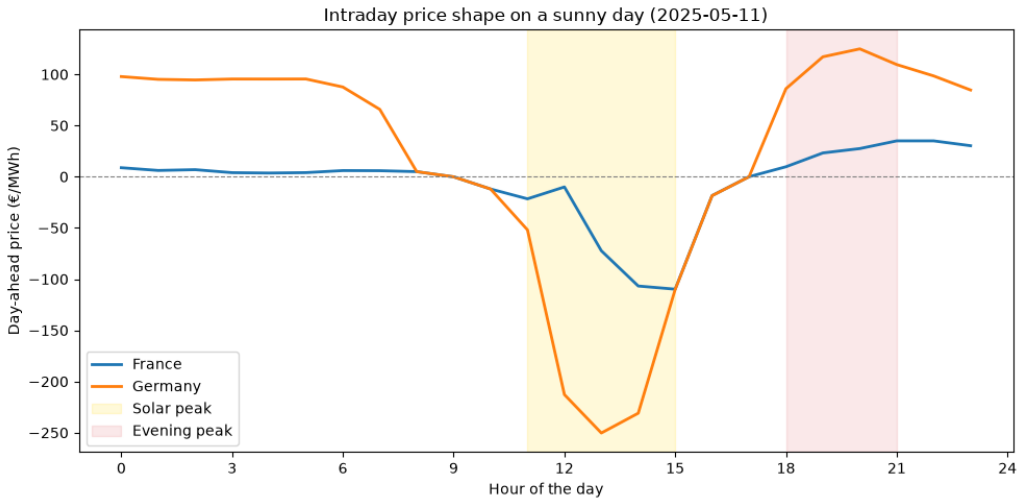


Figure 1: Day-ahead prices in France and Germany on Sunday 11 May 2025. Source: Energy-Charts (Fraunhofer ISE), EPEX Spot day-ahead prices.

This first shape is very consistent with the generation mix of each country. On a sunny day, during the solar peak (11h–15h), electricity in Germany is cheaper, as solar accounts for a much larger share of the German generation mix than of the French one. But when the sun sets and solar panels can no longer produce electricity, the system has to rely on other technologies to keep meeting demand. We can see that German prices explode in the evening, while French prices stay lower than the German ones, as nuclear power continues to cover most of the French demand at a low marginal cost.

When looking at this graph with a battery buying low and selling high in mind, the answer seems obvious: we clearly want to install our battery in Germany. But this graph only represents one day, which happens to be a Sunday with low demand, and it may not be representative. To resolve this problem, I tried to be more precise with an average over the entire month.

1.4 Average price profile over a month

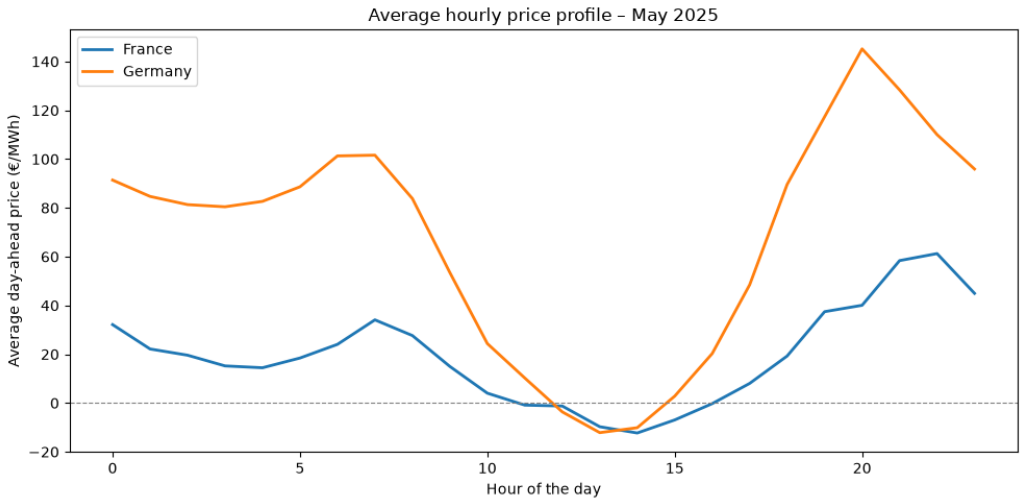


Figure 2: Average hourly day-ahead price profile in France and Germany, May 2025. Source: Energy-Charts (Fraunhofer ISE), EPEX Spot day-ahead prices.

The monthly average confirms what we observed on a single day. Even if prices during the solar peak are almost the same in both countries, around or slightly below zero, prices in the evening and at night are much higher in Germany, and a battery in Germany would likely be far more profitable. On average in May 2025, the gap between the cheapest and the most expensive hour is about 155 €/MWh in Germany, against about 75 €/MWh in France.

To compare the results across years, we average the prices hour by hour over every day of May, to get a single profile for May 2019, May 2022, May 2024 and May 2025. To keep the comparison readable, we do not plot every year but select four years that each represent a distinct market regime:

- 2019 is the baseline, before the solar boom and the energy crisis.
- 2022 is the peak of the energy crisis, with record gas prices and reduced French nuclear availability.
- 2024 and 2025 reflect the current market, with a high share of solar energy in both countries.

The other years are left out of this comparison because they mainly reflect exceptional or transitional conditions. In 2020, electricity demand dropped sharply during the COVID-19 lockdowns, which distorted prices. 2021 and 2023 are transition years, respectively into and out of the energy crisis: their prices mix several regimes and add little to the comparison. However, all years from 2019 to 2025 are included in the battery simulation, so that no period is excluded from the final results.

1.5 How the daily price shape has evolved since 2019

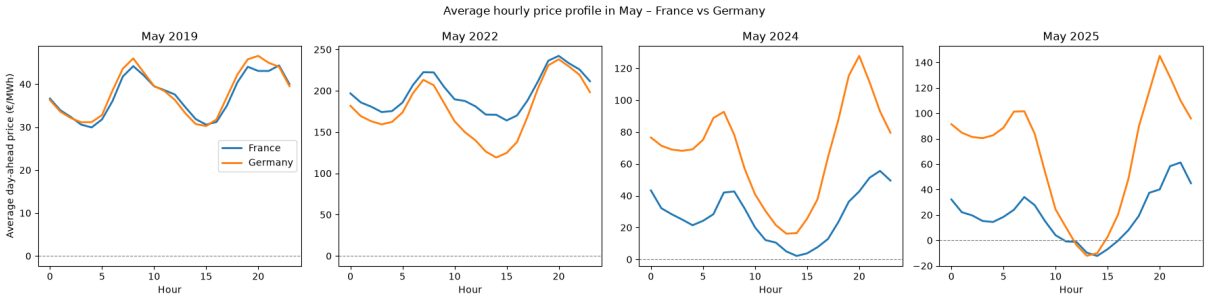


Figure 3: Average hourly day-ahead price profile in May, France vs Germany (2019, 2022, 2024, 2025). Source: Energy-Charts (Fraunhofer ISE), EPEX Spot day-ahead prices.

Table 1: Average daily spread (maximum minus minimum hourly price of each day) in May, in €/MWh.

Year	France	Germany	DE/FR ratio
2019	23.4	25.0	1.07
2022	96.0	132.3	1.38
2024	64.0	116.3	1.82
2025	83.3	161.6	1.94

A trend seems to be appearing. The average daily spread, which is what a battery seeks to capture, has grown in both countries, but much faster in Germany: the ratio between the German and the French spread rose from 1.07 in 2019 to 1.94 in 2025. Whereas prices in France and Germany were almost the same at every hour of the day in 2019, with a small daily spread, we can see an increasing gap between the two countries. 2022 stands out as an exception, with French prices above German ones due to reduced nuclear availability. In 2024 and 2025, the gap grows particularly outside the solar peak: German electricity prices are not exceptionally low during the solar peak, where they are close to French prices, but they are exceptionally high during the other hours of the day.

2 The battery simulation

To simulate our battery on the electricity market, we have to define its optimisation function and the physical constraints associated with the battery.

2.1 Model and assumptions

We simulate a 1 MW / 2 MWh battery that trades on the day-ahead market. For each day, we solve a linear optimisation problem under constraints: knowing the hourly prices p_t, the battery chooses when to charge and discharge in order to maximise its daily profit.

Decision variables (for each hour t):

- c_t: charging power (MW)
- d_t: discharging power (MW)
- s_t: state of charge at the end of hour t (MWh)

To model our optimisation function, we want to maximise our profit, represented by $d_t \times p_t$, because every time we discharge power at time t, we sell it at the price p_t. With the same method, we need to subtract $c_t \times p_t$, as when we charge power at time t, we buy it at the price p_t. Finally, we need to represent the degradation of our battery.

Finally, we have our final objective: maximise the daily profit represented by:

$$\max \sum_t \left[p_t \left(d_t - c_t \right) - k \, d_t \right] \tag{1}$$

where k is a degradation cost per MWh discharged that we will determine later.

Constraints:

- Power limit: $0 \leq c_t \leq P$ and $0 \leq d_t \leq P$, with $P = 1$ MW, because the battery can't charge or discharge faster than its rated power P.
- Capacity limit: $0 \leq s_t \leq E$, with $E = 2$ MWh the capacity.
- Energy balance: $s_t = s_{t-1} + \eta \, c_t - \dfrac{d_t}{\eta}$, with $\eta = \sqrt{0.85}$ (85% round-trip efficiency, source: Cole et al. [3]). The state of charge at the end of period t is the state of charge at the beginning plus what we add (we multiply by η to consider the efficiency) minus what we take out (we divide by η).
- The battery starts and ends each day half full: $s_0 = s_T = E/2$.
- Charging + discharging power: $c_t + d_t \leq P$. Without this constraint, the optimiser could charge and discharge in the same hour during negative prices, using round-trip losses to absorb extra energy. Adding it changes total revenues by about 0.1%.

Assumptions:

- Perfect foresight: the battery knows all prices of the day in advance.
- Price taker: the battery is too small to move market prices.
- Only day-ahead arbitrage is modelled, without intraday trading or ancillary services.

Determination of our factor k: k depends on the price of the battery, as it depends on how much it would cost us to replace it. To try being as realistic as possible in our model, we can't take a constant k, because the price of the battery between 2019 and 2025 is (very) different.

Degradation cost k Each MWh discharged wears the battery a little. We price this degradation as the replacement cost of the battery spread over its cycle life:

$$k = \frac{\text{battery pack price (€/MWh of capacity)}}{\text{cycle life (number of full cycles)}} \tag{2}$$

Level in 2025: BloombergNEF reports an average pack price of 70 \$/kWh for stationary storage in 2025 [4]. With a conservative cycle life of 6,000 full cycles for LFP storage cells (CATL announces up to 10,000 [6]), this gives $k \approx 70\,000/6\,000 \approx 12$ \$/MWh, i.e. about 10 €/MWh.

Evolution since 2019: Battery prices have fallen sharply since 2019, so the degradation cost was higher in earlier years. We scale k with BloombergNEF's annual average pack price (all segments), which is the only public series covering the whole period:

$$k_y = k_{2025} \times \frac{\text{pack price}_y}{\text{pack price}_{2025}} \tag{3}$$

Table 2: Average battery pack price and resulting degradation cost k. Source: BloombergNEF, annual Lithium-Ion Battery Price Survey (press releases, 2019–2025) [5].

Year	Average pack price ($/kWh)	k (€/MWh)
2019	156	14.4
2020	137	12.7
2021	132	12.2
2022	151	14.0
2023	139	12.9
2024	115	10.6
2025	108	10.0

2.2 Price data, 2019–2025

We download hourly day-ahead prices for France and Germany-Luxembourg from 1 January 2019 to 31 December 2025 from Energy-Charts (Fraunhofer ISE) [1]. Data are retrieved month by month and cached locally to respect the API rate limit. Since October 2025, day-ahead prices are published at a 15-minute resolution; they are averaged to hourly values so that all years are comparable (see the code in Appendix A). Our data is collected and has no missing values, so we can continue with the optimisation. To optimise our function, we use `linprog` (SciPy) with the HiGHS method.

2.3 Annual arbitrage revenues

To test our model, we first run it on the sunny day used earlier (Sunday 11 May 2025). The battery earns 256 € in France and 726 € in Germany. In both countries, most of the gain comes from being paid to charge during the negative prices of the afternoon. The German battery also earns more by discharging at high prices in the morning and in the evening. This single day is consistent with the price shapes observed earlier, but the full simulation is needed to draw conclusions.

To continue, we will run the simulation on our 7 years (from 2019 to 2025). As we said earlier, we will adapt k depending on the year.

Table 3: Annual day-ahead arbitrage revenue of a 1 MW / 2 MWh battery (k€/MW/year), with k by year.

Year	France	Germany	DE/FR ratio
2019	3.5	5.7	1.63
2020	4.7	8.2	1.74
2021	26.4	33.2	1.26
2022	69.2	92.2	1.33
2023	35.5	44.9	1.26
2024	38.4	59.2	1.54
2025	48.4	68.6	1.42

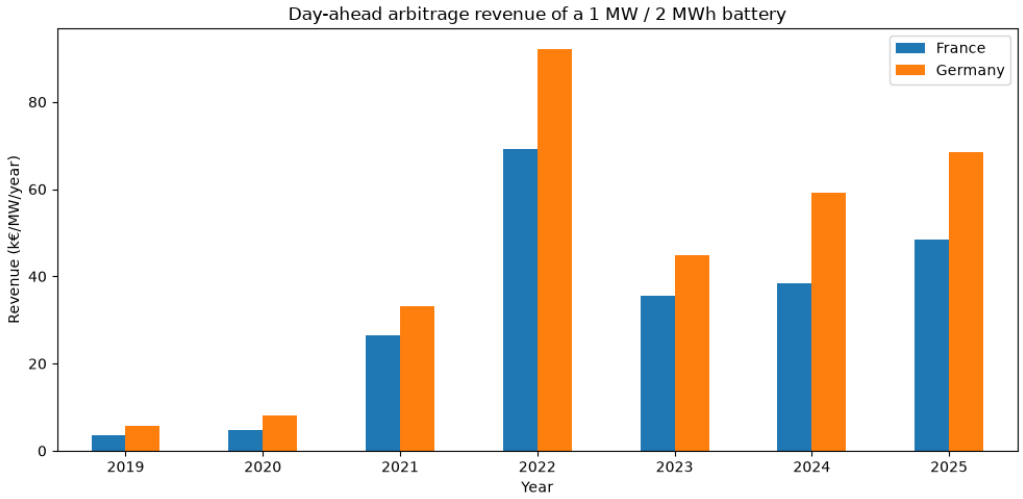


Figure 4: Annual day-ahead arbitrage revenue of a 1 MW / 2 MWh battery, France vs Germany.

We can see that between 2019 and 2025, day-ahead arbitrage revenues rose more than ten times in both countries. A battery consistently earns more in Germany than in France, by 26% to 74% depending on the year, but the gap has not widened very strongly: the divergence observed in May is mostly a spring effect, driven by the high share of solar energy in the German energy mix. To confirm that this gap between our two batteries comes mostly from the sunny months, we calculate the ratio for each month of each year. Logically, we should observe larger ratios during sunny months.

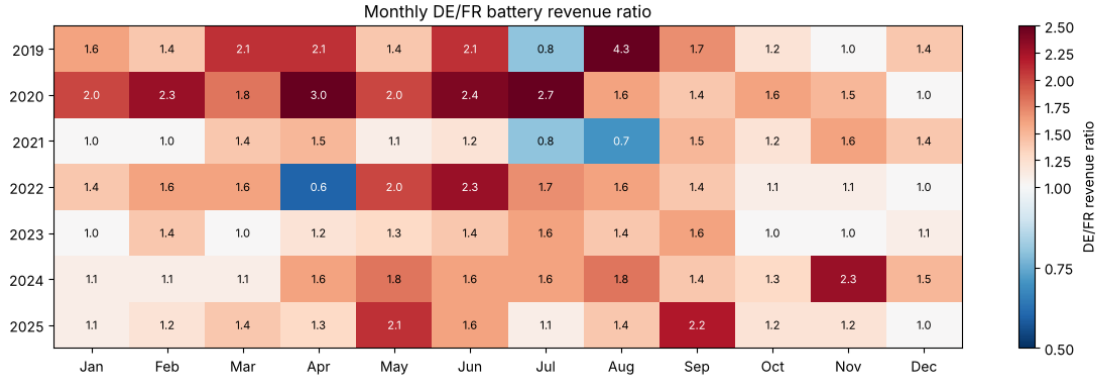


Figure 5: Monthly ratio between German and French battery revenues (DE/FR), 2019–2025.

Since 2024, the German advantage is mostly concentrated between April and September, when solar output is high (ratios of 1.1 to 2.2), while winter months usually show similar revenues in both countries (ratios of 1.0 to 1.5). The main exception is November 2024 (ratio of 2.3), driven by the Dunkelflaute episodes, when Germany had neither wind nor sun. In earlier years, the monthly ratios are noisier: revenues were small in 2019–2020, and the 2021–2023 energy crisis, including the French nuclear problems of 2022, distorted the comparison.

As shown by Kumtepeli et al. [7], our degradation factor k has a major role in the determination of the profit, and a small change in k can totally modify the output of our tests. To test the robustness of our study, we perform a sensitivity analysis to compare the results we have with different values of k.

Table 4: Sensitivity of annual revenue (k€/MW/year) and of the DE/FR ratio to the degradation cost k (€/MWh).

	$k = 5$			$k = 20$			k by year		
Year	FR	DE	Ratio	FR	DE	Ratio	FR	DE	Ratio
2019	8.5	11.4	1.34	2.0	3.9	1.95	3.5	5.7	1.63
2020	9.0	13.6	1.51	2.7	5.2	1.93	4.7	8.2	1.74
2021	33.0	39.8	1.21	21.0	27.5	1.31	26.4	33.2	1.26
2022	78.2	102.0	1.30	63.6	86.1	1.35	69.2	92.2	1.33
2023	44.0	53.2	1.21	29.0	38.3	1.32	35.5	44.9	1.26
2024	44.9	65.5	1.46	29.7	50.3	1.69	38.4	59.2	1.54
2025	54.5	74.0	1.36	38.4	59.2	1.54	48.4	68.6	1.42

As we could predict, revenues are highly impacted by the choice of k. But we can also tell that our results are robust to the choice of degradation cost: in every case, the German battery is more profitable than the French one. Now that we have noticed this result, we can ask ourselves the reasons for such large differences between the two countries.

3 What drives the gap between France and Germany?

We will try to analyse the energy production mix of both countries to better understand where our results come from. For that, we download public generation data for both countries over our 2019–2025 period.

3.1 Generation data, 2019–2025

We download the net electricity production of each source (solar, wind, nuclear, gas…) and the electricity consumption (load) of both countries from Energy-Charts [1], month by month, in the same way as the prices. For each day, we compute the share of the daily consumption covered by solar, wind and nuclear power. Missing values are set to zero for generation sources, as they correspond to no production (solar at night, French offshore wind before its first wind farms, German nuclear power after April 2023). Only two days remain incomplete in France and are excluded.

Table 5: Average daily share of load covered by each source (%). Source: Energy-Charts (Fraunhofer ISE).

	Solar		Wind		Nuclear	
Year	FR	DE	FR	DE	FR	DE
2019	2.7	8.7	7.1	24.8	82.1	14.4
2020	3.1	9.9	8.5	26.3	75.4	12.7
2021	3.2	9.4	7.6	22.4	79.1	13.1
2022	4.4	11.7	8.4	25.0	62.4	6.9
2023	5.5	12.4	11.4	30.2	76.1	1.4
2024	5.9	13.5	10.6	28.9	84.7	0.0
2025	7.5	15.9	11.2	28.0	86.2	0.0

To visualise the data more easily, we put it in a graph:

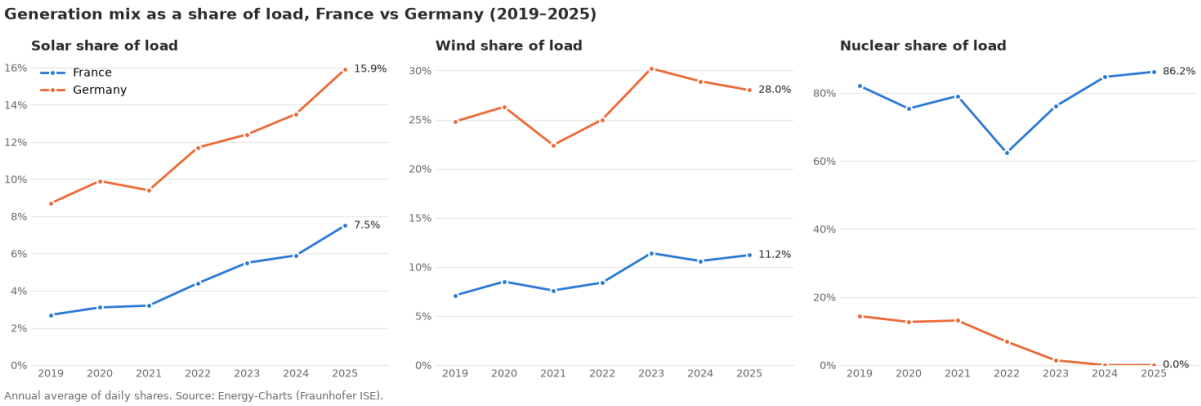


Figure 6: Generation mix as a share of load, France vs Germany (2019–2025).

3.2 Solar share and battery revenues

The chart above shows how much of each country's electricity consumption is covered by solar, wind and nuclear power, on average, for each year between 2019 and 2025.

Solar: Germany produces much more solar electricity than France: in 2025, solar covers about 16% of German consumption, against 7.5% in France. Solar is growing in both countries, and it is growing faster in France in relative terms (almost three times more than in 2019), but France is still far behind Germany.

Wind: The same pattern appears for wind: about 28% of German consumption in 2025, against 11% in France.

Nuclear: This is where the two countries are completely different. In France, nuclear power produces the equivalent of 75% to 86% of the country's consumption every year, except in 2022, when many reactors were stopped for maintenance and corrosion problems. In Germany, nuclear power slowly disappeared and reached zero after the last reactors were closed in April 2023.

These numbers are consistent with the price shapes we observed. Germany has more solar, so its midday prices fall lower. But the most important difference is in the evening: when the sun sets, Germany has no nuclear power left and relies more on gas plants, which are expensive and set a high price. France still has a large amount of cheap nuclear power, which keeps its evening prices low. A battery earns money from the gap between midday and evening prices, so this gap is larger in Germany.

3.3 Regression analysis

To add the gas price to our analysis, we use the European natural gas price published monthly by the International Monetary Fund (based on the Dutch TTF hub) [8], converted from USD/MMBtu to USD/MWh. Each day receives the price of its month.

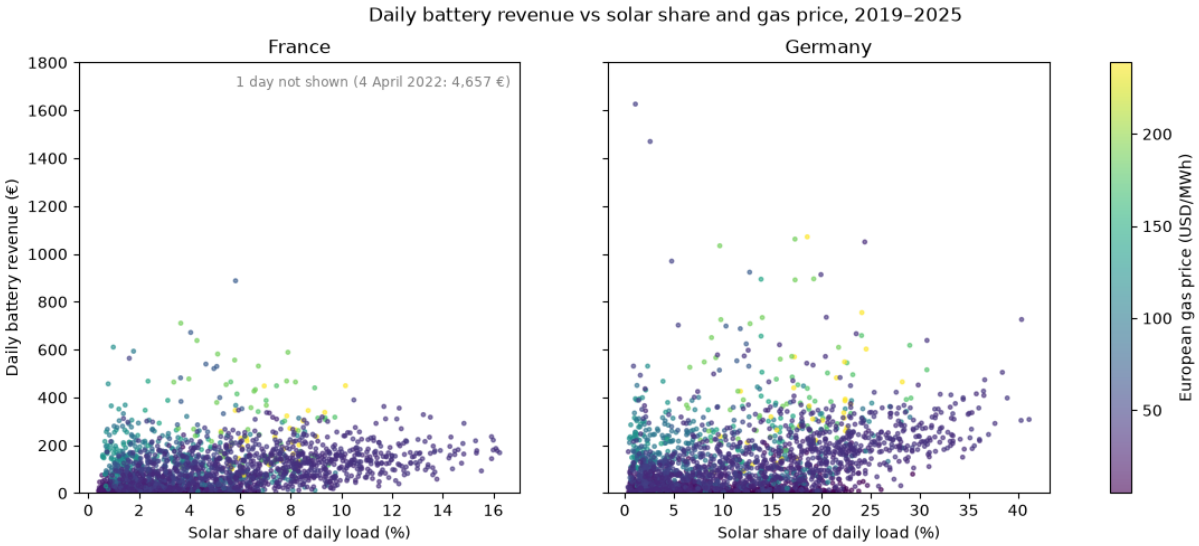


Figure 7: Daily battery revenue vs solar share of daily load, coloured by the European gas price (2019–2025). One French day (4 April 2022, 4,657 €) is above the chart limit. Sources: Energy-Charts (Fraunhofer ISE); IMF via FRED.

We can observe that, in general, at a given solar share of daily load, our daily battery revenue follows European gas prices: when gas prices are high, our daily battery revenue tends to be higher as well. We can also observe that a larger solar share of daily load is associated with a higher daily battery revenue. These two charts confirm what we were saying earlier: a large solar share and high gas prices are both associated with higher daily battery revenues. We need to note that the two countries do not cover the same range: the solar share reaches up to 40% of daily load on some days in Germany, against 16% at most in France. One day is not shown in the French chart (4 April 2022, 4,657 €), as it was flattening the rest of the graph.

To go further than the charts, we run a linear regression for each country. We try to explain the daily battery revenue with the solar share, the wind share and the nuclear share of daily load (in percentage points), and with the European gas price (in USD/MWh). Standard errors are corrected for autocorrelation (Newey-West), as each day tends to look like the day before.

Table 6: Regression of the daily battery revenue (€ per day) on the generation mix and the gas price, 2019–2025.

Variable	France	Germany
Solar share (€ per point)	9.6	4.5
Wind share (€ per point)	2.2	0.1
Nuclear share (€ per point)	−0.4	−5.6
Gas price (€ per USD/MWh)	1.20	1.66
R^2	0.23	0.45

All coefficients are significant at the 1% level, except the wind share in Germany ($p = 0.69$) and the nuclear share in France ($p = 0.51$).

Solar: In both countries, more solar means more revenue for the battery. Surprisingly, one more point of solar adds more revenue in France (9.6 € per day) than in Germany (4.5 € per day). We can explain this by the fact that Germany already has a lot of solar: midday prices are often already very low or negative, so each additional point of solar deepens the midday trough less. Solar cannibalises its own value. In 2025, solar adds about the same amount per day in both countries (around 70 €), so solar alone does not explain the gap between France

and Germany.

Gas: A higher gas price means more revenue in both countries, but the effect is stronger in Germany (1.66 € against 1.20 € per day for each USD/MWh). This is consistent with gas plants setting the evening price more often in Germany, while French nuclear power keeps evening prices low in France. With a gas price of about 40 USD/MWh, this represents around 18 € per day more in Germany, or roughly 6 to 7 k€ per year.

Nuclear: In France, the nuclear share has no significant effect: it is always high and changes little from one day to the next. In Germany, a higher nuclear share is associated with lower revenues. I think that this is because German nuclear power decreased a lot until 2023 while solar and battery revenues were rising, so this coefficient also captures this downward trend over time.

Overall: The regression suggests that the German advantage comes less from solar power than from the evening: German revenues depend more on the gas price, as Germany no longer has nuclear power to keep evening prices low. This matches the most profitable German days, which are winter days with no wind and no sun, not sunny days (the two best German days are 12 December and 6 November 2024).

Limits: We need to be careful with these results: they show that some variables move together, not that one causes the other. Our model also only explains part of the daily revenues: about 23% in France and 45% in Germany. We only have a monthly gas price, so we cannot see how gas prices change from one day to the next. Finally, we use a linear model, which assumes that each extra point of solar always has the same effect, while we saw that this effect gets smaller when there is already a lot of solar.

4 Limitations

Our model is a simplified version of how a real battery works, and our results must be read with this in mind.

The battery model:

- *Perfect foresight:* We assume that the battery knows all the prices of the day in advance. In reality, a battery operator has to submit its bids before the day-ahead auction, without knowing the final prices. Our revenues are therefore an upper bound of what a battery could earn on the day-ahead market.
- *Day-ahead market only:* A real battery also trades on the intraday market and sells ancillary services to the grid operator (frequency reserves), which are often a large part of its revenues. For comparison, I found a source saying that the enervis Battery Storage Index forecast about 134 k€/MW for a 1 MW / 2 MWh battery in Germany in 2025, all markets included [9], while our day-ahead model gives 68.6 k€/MW. Our revenues therefore only show part of what a real battery earns.
- *Price taker:* We assume that our battery is too small to change market prices. This is true for a 1 MW battery, but not for the several gigawatts of batteries now being built in Germany: as more batteries charge at midday and discharge in the evening, they reduce the very price gap they earn money from.

The data:

- Since October 2025, day-ahead prices are published every 15 minutes. We average them to hourly values so that all years are comparable, which slightly reduces the price gaps the battery can capture at the end of 2025.

- We only have a monthly gas price, so we cannot see how gas prices change from one day to the next.

The degradation cost: The cost of battery wear k is uncertain: it depends on battery prices and on how many cycles a battery can really do before it wears out. This is why we tested several values of k. Our main conclusion (a battery earns more in Germany than in France) holds for all of them, but the size of the revenues and of their increase since 2019 depends on this choice.

The regression: Our regression shows correlations, not causes. It only explains part of the daily revenues, and it assumes that each extra point of solar always has the same effect, while this effect gets smaller when there is already a lot of solar. Some variables also move together over time (more solar, less German nuclear power), which makes their effects harder to separate.

5 Conclusion

At the beginning of this project, we wanted to understand why a battery earns less in France than in Germany, and how this gap has evolved since 2019.

Our first look at a single sunny day suggested a clear answer: German prices explode in the evening while French prices stay low, and we clearly wanted to install our battery in Germany. The full simulation over seven years confirms this first intuition, but makes it more precise.

Between 2019 and 2025, day-ahead arbitrage revenues rose more than ten times in both countries: flexibility has become much more valuable on both sides of the border. A battery consistently earns more in Germany than in France, by 26% to 74% depending on the year, and this result holds for every value of the degradation cost we tested. However, the gap has not widened very strongly over time. The strong divergence we observed in May is mostly a spring effect: since 2024, the German advantage is concentrated between April and September, while winter months usually show similar revenues in both countries, except during windless and sunless episodes.

The reason behind this gap is not only solar power. Germany has two to three times more solar than France, which makes its midday prices fall lower, but each extra point of solar adds less and less value as solar becomes abundant. The main difference comes from the evening: when the sun sets, Germany relies on gas plants, which set a high price, while France still has a large amount of cheap nuclear power that keeps its prices low. This is why German battery revenues depend more on the gas price, and why the most profitable German days are winter days with no wind and no sun.

These results only cover the day-ahead market, with a simplified battery that knows prices in advance, so they show part of what a real battery can earn. Still, they show a clear trend: as solar keeps growing in both countries and as Germany relies more on gas in the evening, the value of storage keeps rising, and the French market could become more attractive for batteries as its own solar share continues to grow.

References

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A Code

Price download (Energy-Charts API, with cache and retry)

```python
def get_prices(bzn, start, end, max_retries=5):
    """Day-ahead prices from Energy-Charts, with disk cache and retry on 429."""
    cache_file = CACHE / f"{bzn}_{start}_{end}.csv"
    if cache_file.exists():
        s = pd.read_csv(cache_file, index_col=0, parse_dates=True).squeeze("columns")
        s.index = pd.to_datetime(s.index, utc=True).tz_convert("Europe/Paris")
        return s

    url = "https://api.energy-charts.info/price"
    wait = 10
    for attempt in range(max_retries):
        r = requests.get(url, params={"bzn": bzn, "start": start, "end": end}, timeout=30)
        if r.status_code == 429:
            print(f"429 on {bzn} {start} - waiting {wait}s (attempt {attempt+1}/{max_retries})")
            time.sleep(wait)
            wait *= 2
            continue
        r.raise_for_status()
        data = r.json()
        s = pd.Series(data["price"],
                      index=pd.to_datetime(data["unix_seconds"], unit="s", utc=True),
                      name=bzn).tz_convert("Europe/Paris")
        s.to_csv(cache_file)
        time.sleep(3)
        return s

    raise RuntimeError(f"Still rate-limited after {max_retries} attempts for {bzn} {start}")

def get_range(bidding_zone, start_year, end_year):
```

```
    parts = []
    for y in range(start_year, end_year + 1):
        for m in range (1,13):
            start = pd.Timestamp(year=y, month=m, day=1)
            end = start + pd.offsets.MonthEnd(0)
            parts.append(get_prices(bidding_zone,start.strftime("%Y-%m-%d"), end.strftime("%Y-%m
    -%d")))
    s = pd.concat(parts)
    return s.resample("1h").mean()
```

Battery optimisation

```
P, E, RTE = 1.0, 2.0, 0.85

def optimise_day(prices, k, P=P, E=E, rte=RTE, exclusive=True, return_schedule=False):
    p = np.asarray(prices, dtype=float)
    T = len(p)
    eta = np.sqrt(rte)
    cost = np.concatenate([p, -p + k, np.zeros(T)])

    A_eq, b_eq = [], []
    for t in range(T):
        row = np.zeros(3 * T)
        row[t] = -eta
        row[T + t] = 1 / eta
        row[2 * T + t] = 1
        if t > 0:
            row[2 * T + t - 1] = -1
        A_eq.append(row)
        b_eq.append(E / 2 if t == 0 else 0.0)
    end = np.zeros(3 * T); end[-1] = 1
    A_eq.append(end); b_eq.append(E / 2)

    A_ub, b_ub = None, None
    if exclusive:
        A_ub = np.zeros((T, 3 * T))
        A_ub[np.arange(T), np.arange(T)] = 1
        A_ub[np.arange(T), T + np.arange(T)] = 1
        b_ub = np.full(T, P)

    bounds = [(0, P)] * (2 * T) + [(0, E)] * T
    res = linprog(cost, A_ub=A_ub, b_ub=b_ub, A_eq=np.array(A_eq), b_eq=b_eq,
                  bounds=bounds, method="highs")
    if not res.success:
        return (np.nan, None) if return_schedule else np.nan
    if return_schedule:
        return -res.fun, res.x
    return -res.fun

def daily_revenues(s, k):
    s = s.dropna()
    out = {}
    for day, x in s.groupby(s.index.date):
        if len(x) < 23:
            continue
        k_day = k[day.year] if isinstance(k, dict) else k
        out[day] = optimise_day(x.values, k_day)
    return pd.Series(out)

pack_price = {2019: 156, 2020: 137, 2021: 132, 2022: 151, 2023: 139, 2024: 115, 2025: 108} #from
     BNEF, expressed in $/kWh
k_by_year = {y: 10 * p / pack_price[2025] for y, p in pack_price.items()}
```

Generation mix and regression

```
GEN_COLS = ["Solar", "Nuclear", "Wind onshore", "Wind offshore", "Fossil gas"]

def daily_features(power):
    """Daily energy (MWh) and shares of load."""
    power = power.copy()
    gen = [c for c in GEN_COLS if c in power.columns]
    power[gen] = power[gen].fillna(0)                        # no data = no production (night,
    offshore before 2023, DE nuclear after April 2023)
    daily = power.groupby(power.index.date).sum(min_count=20)
    out = pd.DataFrame(index=daily.index)
    out["solar_share"] = daily["Solar"] / daily["Load"]
    out["wind_share"] = (daily["Wind onshore"] + daily.get("Wind offshore", 0)) / daily["Load"]
    if "Nuclear" in daily.columns:
        out["nuclear_share"] = daily["Nuclear"] / daily["Load"]
    out["load_GWh"] = daily["Load"] / 1000
    return out

def run_regression(df, name):
    X = df[["solar_share", "wind_share", "nuclear_share", "gas"]].copy()
    X[["solar_share", "wind_share", "nuclear_share"]] *= 100 # shares in % so coefficients read
    "€ per percentage point"
    X = sm.add_constant(X)
    model = sm.OLS(df["revenue"], X).fit(cov_type="HAC", cov_kwds={"maxlags": 7})
    print(f"===== {name} =====")
    print(model.summary().tables[1])
    print(f"R²: {model.rsquared:.2f}\n")
    return model
```